Owais Ali Khan

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Education

- **FAST NUCES** — Karachi, Pakistan
  Bachelor of Science in Computer Science; CGPA: 3.88

Experience

- **Securiti AI** — Hybrid
  Associate Software Engineer (Backend) — *June 2025 - Present*
  - Developed and maintained backend **Java** and **Python** services detecting *PII/sensitive data* in unstructured files and databases at enterprise scale.
  - Engineered *file-format* extraction and **OCR** across PDF, MS Office, OpenDocument and email formats, hardening document *parsers* against malformed and adversarial inputs.
  - Built and deployed a new **gunicorn**-based **Python** microservice with multi-stage **Docker** builds, boosting detection-pipeline throughput and scalability.
  - Re-architected state storage on **Redis** and hardened **Kubernetes** workloads (non-root, read-only FS, dropped capabilities); fixed **ReDoS** flaws, drove **CVE remediation**, and root-caused production **OOMs** and pod crash-loops under strict *SLAs*.
- **Securiti AI** — Hybrid
  Software Engineer Intern — *June 2024 - Aug 2024*
  - Engineered high-performance infrastructure for large-scale *regular-expression* matching using **Intel Hyperscan**, achieving speedups of 2–3 orders of magnitude.
- **Retrocausal AI** — Remote
  Applied Research Intern — *Mar 2024 - June 2024*
  - Developed *RESTful* API endpoints and automated reporting workflows using the **Django REST Framework**, with scalable storage and retrieval on **Amazon S3**.
- **LDP Logistics** — Onsite
  IT Support Specialist — *June 2023 - Sep 2023*
  - Built and deployed a full-stack *web application* (**Django**, **Bootstrap4**) for managing electronic *inventory* across local and overseas offices.

Projects

- **LLM Text Detector** — Final Year Project
  Natural Language Processing — *Fall 2024, Spring 2025*
  - **A Hybrid Approach to LLM-generated Text Detection**: Combined *cross-perplexity* scores with a fine-tuned *T5* transformer to detect AI-generated text.
  - Achieved a 94% **F1-score** on a curated dataset (7500 generations per class).
- **Polaris** — Parallel and Distributed Computing
  High-performance Image Processing — *Fall 2023*
  - A *parallelized* image-enhancement program implementing **Gaussian filtering** and **Histogram Equalization**.
- **LDP Asset Manager** — Full Stack Development
  Fullstack Web Application — *Summer 2023*
  - An inventory-management *web portal* (**Django**, **MySQL**, **MUI**) tracking assets, accessories and licenses across all LDP offices.

Skills

- **Languages**: Java, Python, C/C++, Go, Bash, SQL
- **Backend & ML**: Django, FastAPI, Flask, Sklearn, Tensorflow
- **Databases & Messaging**: PostgreSQL, MySQL, Redis, Elasticsearch, Kafka
- **DevOps**: Docker, Kubernetes, Jenkins, NGINX, Git, Linux
- **Cloud**: AWS (S3, ECS, EKS, Cloudwatch, RDS, Elasticache), Azure (AKS, Azure Blob/File Storage)
- **Security**: Trivy, Grype, Aquasec, Sonarqube

Achievements

- **ICPC Asia Topi Regional**: Gold Medallist '24-25 (National 4th rank), Silver Medallist '23-24 (National 8th rank)
- **IEEE Xtreme 17.0**: Global Rank 531/4200
- **Academic**: Rector's List (Spring '22), Dean's List (7 semesters, Fall '21 – Spring '25)

Volunteering & Communities

- **Student Lab Assistant, Data Structures and Algorithms**: Fall '24, Fall '25
- **Mentoring & Events**: ACM Devday'25 Mentor (Jan '25 – Apr '25), Procom'24 Co-head CS Competitions (Jan '24 – Mar '24)
- **IEEE Membership**: Dec '23 – Dec '24